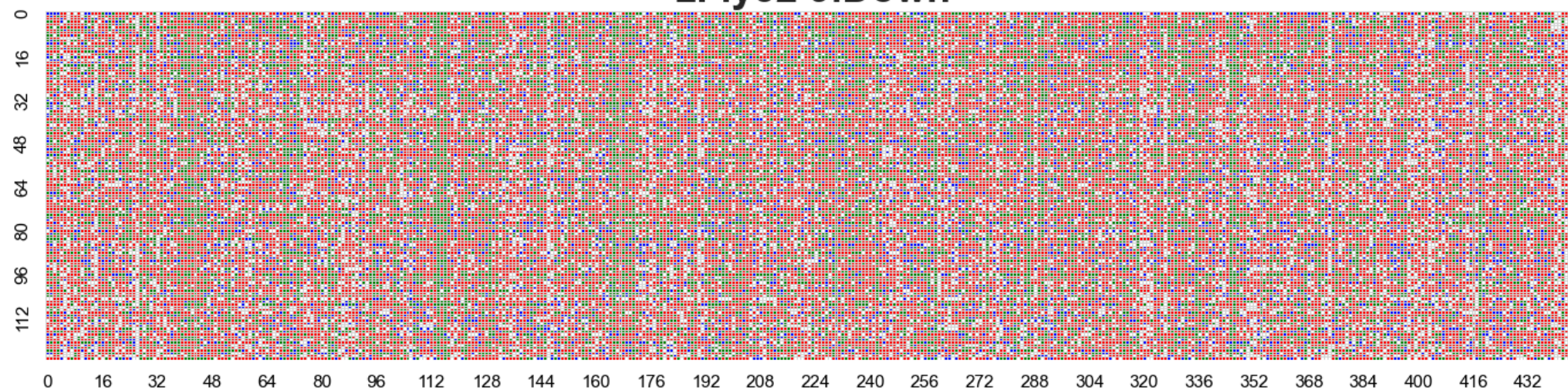
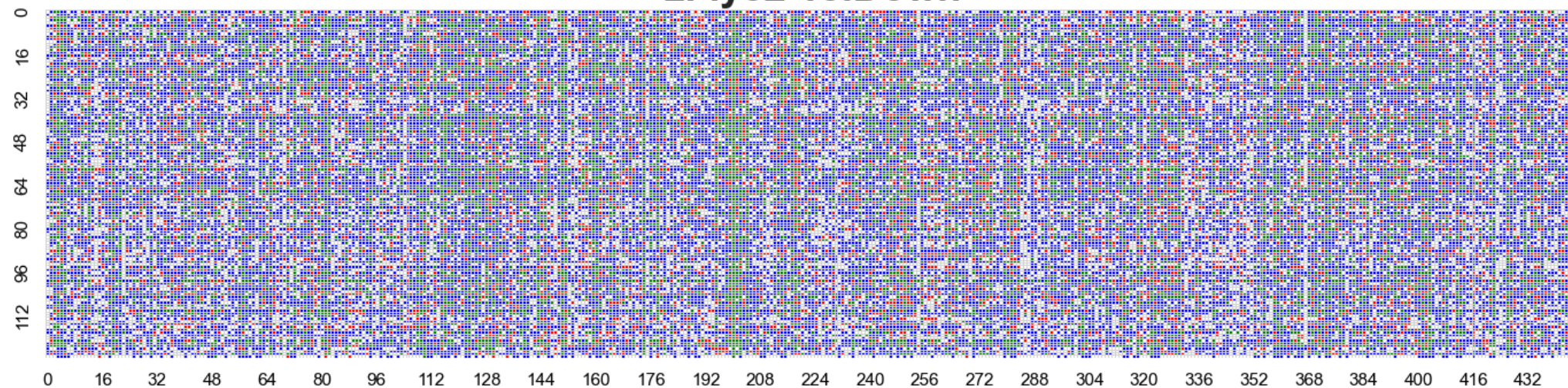


Biased Matrices Are Preserved in Zephyr under 32×32 Block Mutagenesis

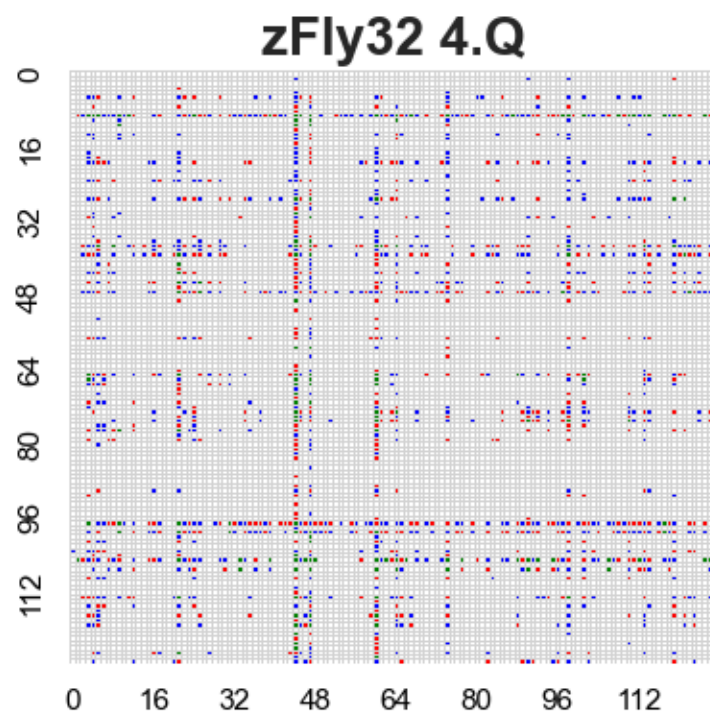
zFly32 9.Down



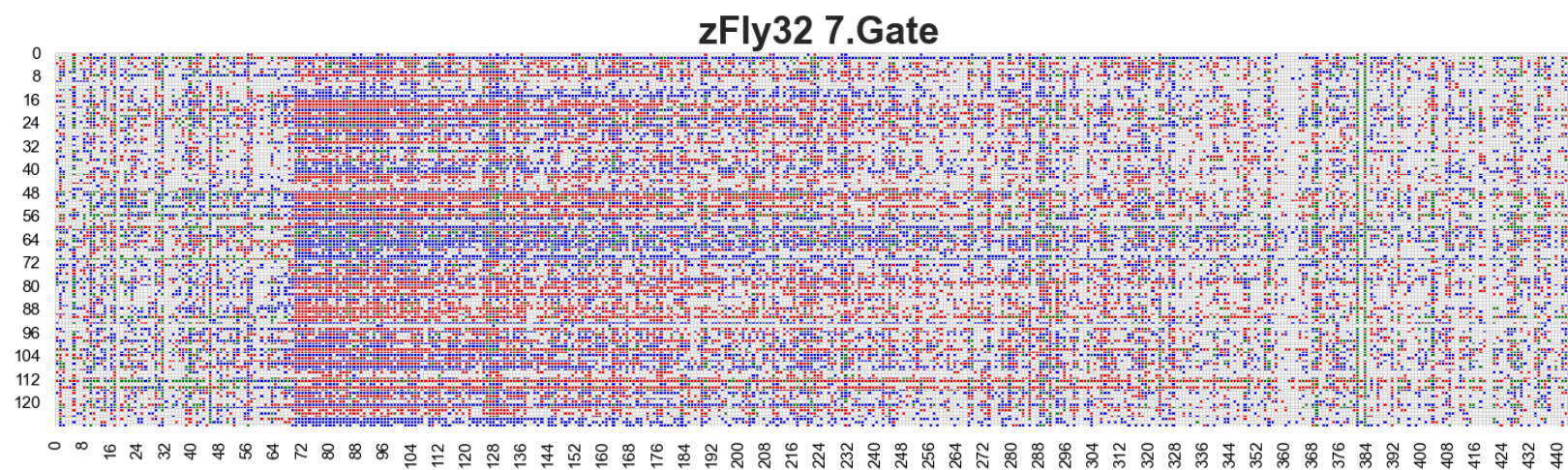
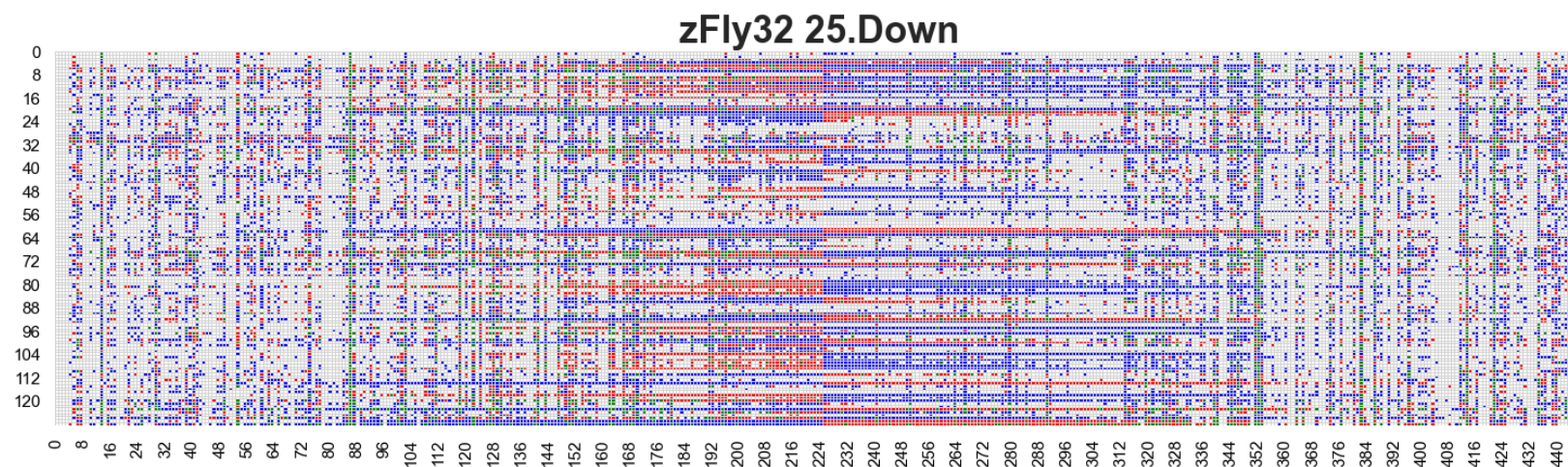
zFly32 13.Down



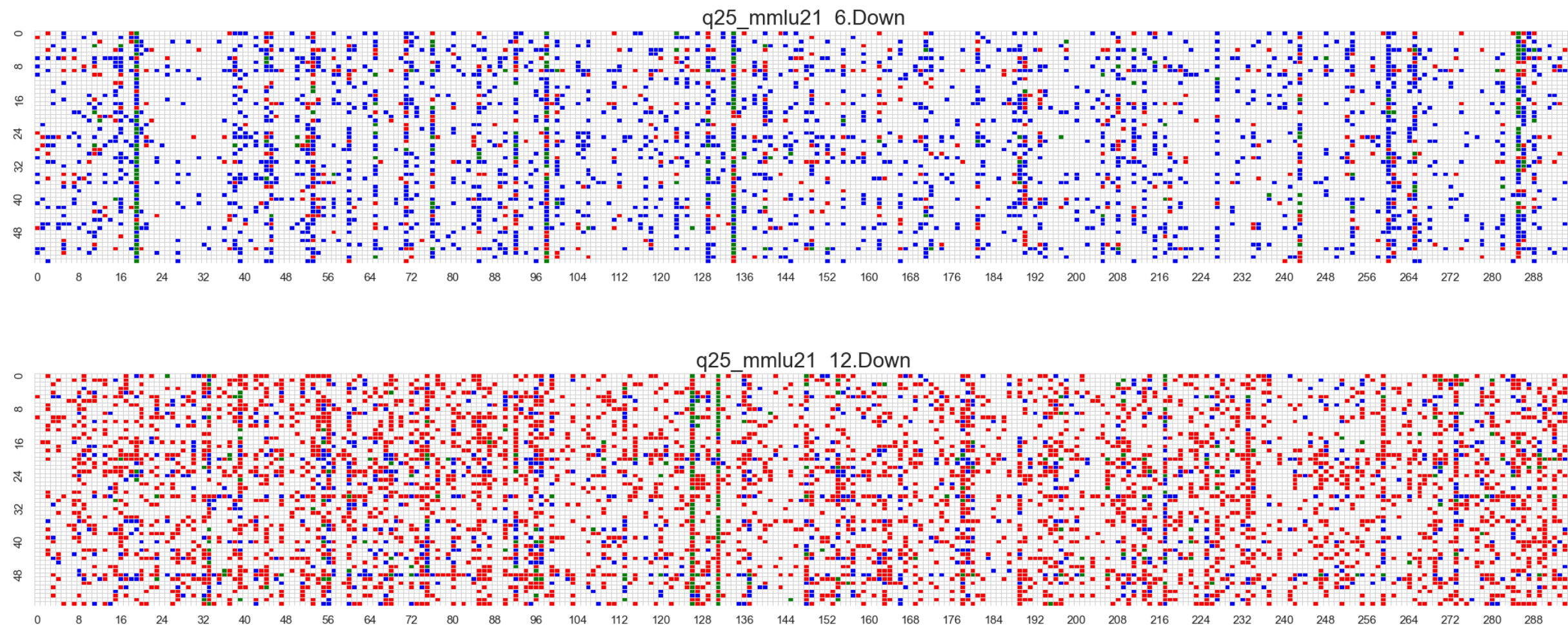
Axial Enrichment Is Preserved in Zephyr under 32×32 Block Mutagenesis



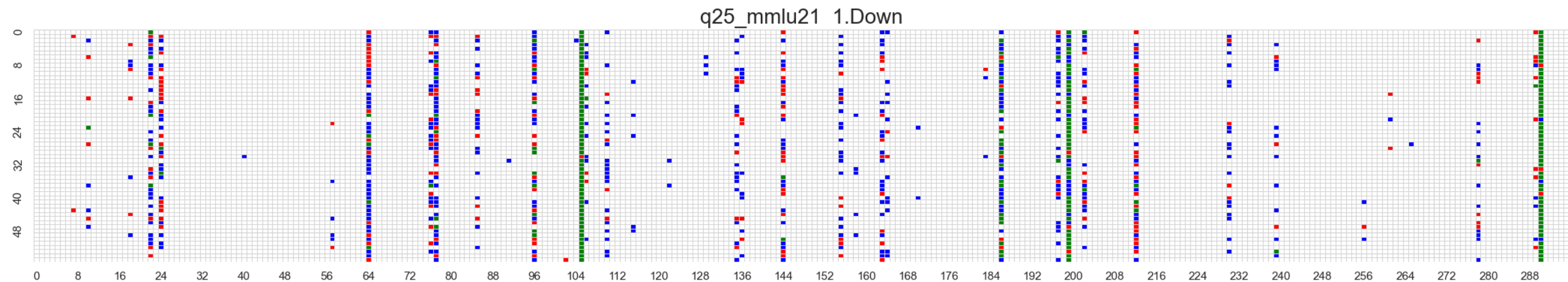
Correlative Complementary Patterns (COPA) Patterns Are Preserved in Zephyr under 32×32 Block Mutagenesis



Biased Matrices Are Preserved in Qwen2.5 under 64×64 Block Mutagenesis

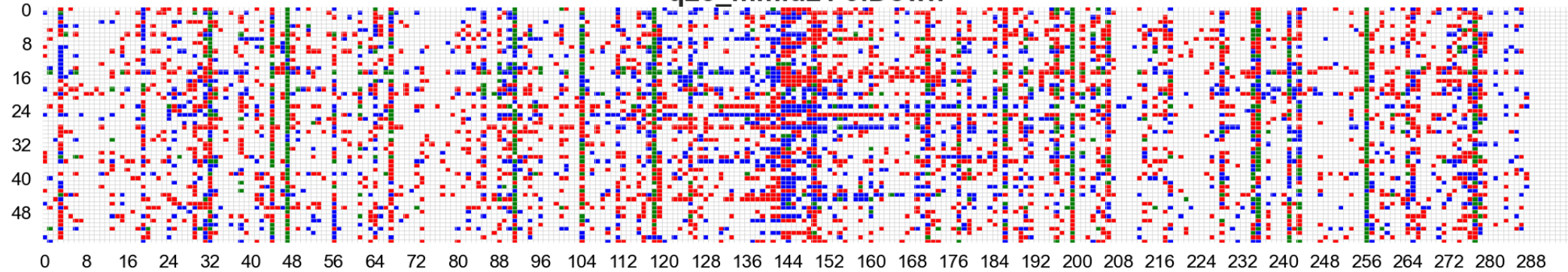


Axial Enrichment Is Preserved in Qwen2.5 under 64×64 Block Mutagenesis

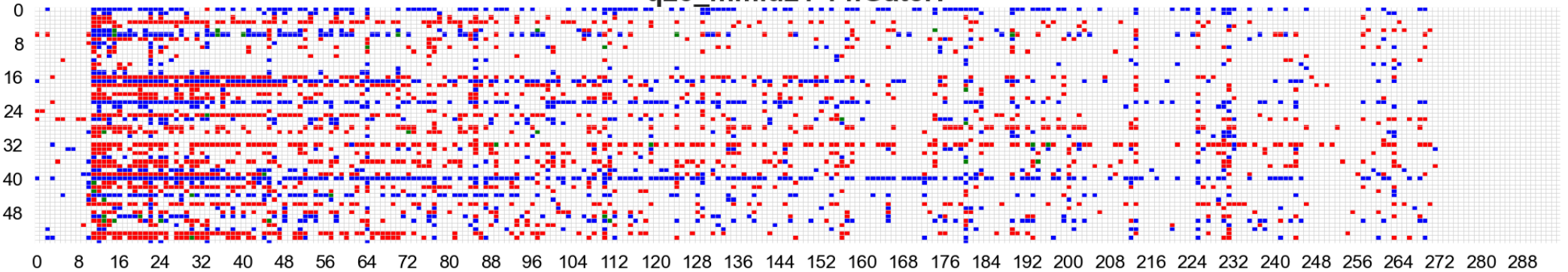


COPA Patterns Are Preserved in Qwen2.5 under 64×64 Block Mutagenesis

q25_mmlu21 9.Down



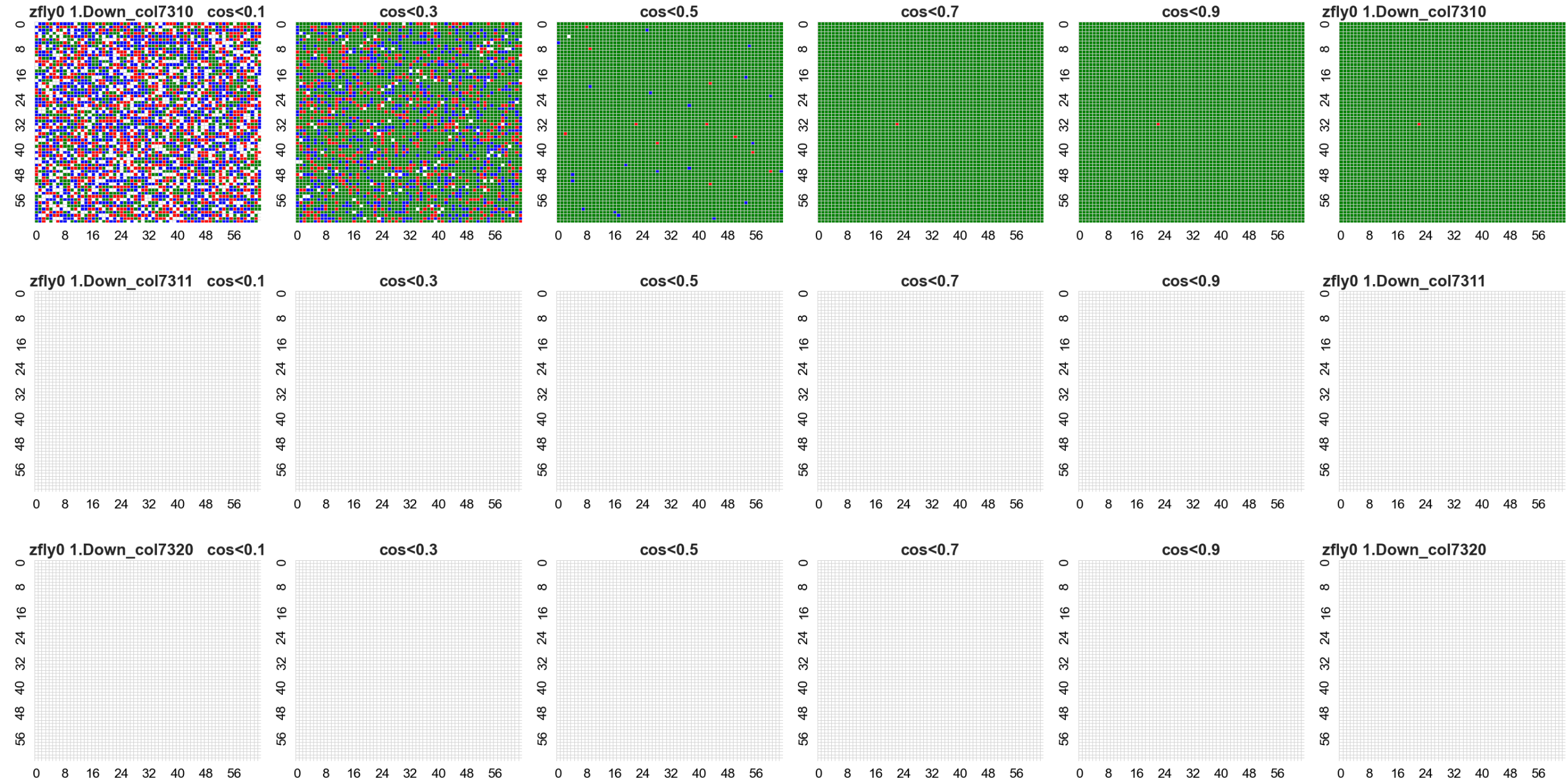
q25_mmlu21 14.Gate.T



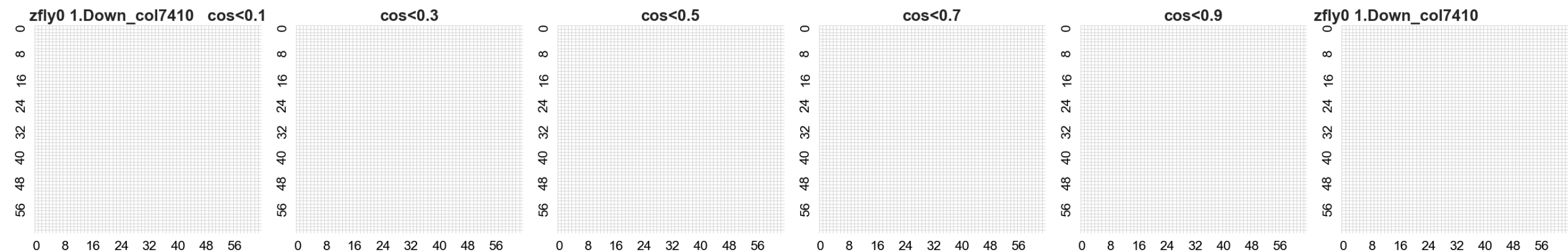
Column-Level Mutagenesis Identifies Column 7310 as a Localized High-Sensitivity Region

- 1×1 mutagenesis was performed on Columns 7310, 7311, 7320, and 7410 of the Zephyr layer-1 MLP down-projection matrix.
- Each column contains 4,063 parameters, which were rearranged into a 64×64 grid for visualization.
- Column 7310 showed severe and recurrent NSMs, whereas nearby and distal control columns showed little or no mutation effect.
- From left to right, NSM maps are shown after applying cosine-similarity filters with thresholds of 0.1, 0.3, 0.5, 0.7, and 0.9, followed by the unfiltered map.
- **Mutation-strength analysis slide (top to bottom):** Column 7310 was mutated using replacement values scaled to 1/2, 1/4, and 1/8 of the original maximum or minimum value. These conditions are denoted as (r_2), (r_4), and (r_8), respectively.

Column-Level Mutagenesis of Zephyr Layer-1 Down Projection: Column 7310 versus Control Columns



Column-Level Mutagenesis of Zephyr Layer-1 Down Projection: Column 7310 versus Control Columns



NSM Maps of Zephyr Column 7310 at Reduced Mutation Strengths

